MATHEMATICS

MATRICES

THEORY SHEET

ENGLISH

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1.1 DEFINITION OF MATRIX

A rectangular array of mn numbers (which may be **real or complex**) in the form of 'm' horizontal lines (called **rows**) and 'n' vertical lines (called **columns**), is called a matrix of order m by n, written as m × n matrix. Such an array is enclosed by [ ] or ( ).

An m × n matrix is usually written as $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$

The above matrix is represented by $A=[a_{ij}]_{m\times n}$.

The number a_{11}, a_{12},.....etc are known as the elements of the matrix A, a_{ij} belongs to the ith row and j^{th} column and is called the **$(i, j)^{th}$** element of the matrix $A=[a_{ij}]$.

E.g., $A=\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 9 \end{bmatrix}$ is a matrix having 2 rows and 3 columns. Its order is 2 × 3 and it has 6 elements :

$a_{11} = 1$, $a_{12} = 2$, $a_{13} = 3$, $a_{21} = 0$, $a_{22} = -1$, $a_{23} = 9$.

Q.1 Construct a 3 × 2 matrix whose elements are given by $a_{ij} = \frac{1}{2}|\, i-3j \,|$.

Ans $\begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$

Sol. In general a 3 × 2 matrix is given by $A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$.

$a_{ij} = \frac{1}{2}|\, i-3j \,|$, i = 1, 2, 3 and j = 1, 2

Therefore $a_{11} = \frac{1}{2}|\, 1-3\times 1 \,| = 1$ $\qquad$ $a_{12} = \frac{1}{2}|\, 1-3\times 2 \,| = \frac{5}{2}$

$a_{21} = \frac{1}{2}|\, 2-3\times 1 \,| = \frac{1}{2}$ $\qquad$ $a_{22} = \frac{1}{2}|\, 2-3\times 2 \,| = 2$

$a_{31} = \frac{1}{2}|\, 3-3\times 1 \,| = 0$ $\qquad$ $a_{32} = \frac{1}{2}|\, 3-3\times 2 \,| = \frac{3}{2}$

Hence the required matrix is given hy $A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$

Q.2 A matrix has 24 elements find total number of different possible orders it can have.

Ans. 8

Sol. $24 \times 1, 1 \times 24, 2 \times 12, 12 \times 2, 3 \times 8, 8 \times 3, 4 \times 6, 6 \times 4$

1.2 TYPES OF MATRICES

Row matrix :

If in a Matrix, there is only one row, then it is called a row matrix (or row vector).

Thus A $[a_{ij}]_{m\,x\,n}$ is a row matrix if m = 1.

eg. [1, 3, 5] is a row matrix of order 1×3.

Column matrix :

A matrix having only one column is called as column matrix (or column vector).

Column matrix is in the form $A = \begin{bmatrix} a_{11} \\ a_{21} \\ \ldots \\ a_{m1} \end{bmatrix}$

Zero matrix :

$A = [a_{ij}]_{m \times n}$ is called a zero matrix, if $a_{ij} = 0\ \forall$ i & j. It is generally denoted by O.

e.g. : (i) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Submatrix : Let A be a given matrix. The matrix obtained by deleting some rows or columns of A is called as submatrix of A.

eg. $A = \begin{bmatrix} a & b & c & d \\ x & y & z & w \\ p & q & r & s \end{bmatrix}$ Then $\begin{bmatrix} a & c \\ x & z \\ p & r \end{bmatrix}, \begin{bmatrix} a & b & d \\ p & q & s \end{bmatrix}, \begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix}$ are all submatrices of A.

Horizontal Matrix :

A matrix of order m × n is a horizontal matrix if **n > m** e.g. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$

Vertical Matrix :

A matrix of order $m \times n$ is a vertical matrix if $\mathbf{m > n}$ e.g. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

Square matrix :

A matrix in which number of rows & columns are equal is called a square matrix. The general form of a square matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ ..... & ..... & ..... & ..... \\ a_{n1} & a_{n2} & ..... & a_{nn} \end{bmatrix}$$ which we denote as $A = [a_{ij}]_n$.

This is a matrix of order "$n \times n$" (or a square matrix of order n)

Important Points for a Square Matrix

1. The pair of elements a_{ij} & a_{ji} are called Conjugate Elements.
2. Elements a_{ii}, $i = 1, 2, ..., n$ are called elements of principal diagonal.
3. The sum of the elements of a **square matrix** A lying along the principal diagonal is called the trace of A i.e. (tr(A)).

   Thus, if $A = [a_{ij}]_{n \times n}$, then $tr(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + ...a_{nn}$

4. For every square matrix a corresponding determinant can be defined

   e.g. - $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad$ $\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

   $B = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ So $\det(B) = |B| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

5. If $A = O$, then $|A| = 0$, but if $|A| = 0$, then A may or may not be zero.

   e.g. $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \Rightarrow |A| = 0$

6. $|A| = 0 \Rightarrow$ A is singular/non-invertible

   $|A| \neq 0 \Rightarrow$ A is non-singular/invertible

1.3 SPECIAL TYPES OF SQUARE MATRICES

Diagonal matrix :

A square matrix $[a_{ij}]_n$ is said to be a diagonal matrix if $a_{ij} = 0$ for $i \neq j$. (i.e., all the elements of the square matrix other than diagonal elements are zero)

Note : Diagonal matrix of order n is denoted as Diag $(a_{11}, a_{22}, ...a_{nn})$.

e.g. : (i) $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ $\quad$ (ii) $\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c \end{bmatrix}$

eg. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ is a diagonal Matrix of order 3 × 3, which also can be denoted by diag [2, 3, 4]

Scalar matrix :

Scalar matrix is a diagonal matrix in which all the diagonal elements are same. $A = [a_{ij}]_n$ is a scalar matrix, if (i) $a_{ij} = 0$ for $i \neq j$ and (ii) $a_{ij} = k$ for $i = j$.

e.g. : (i) $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ (ii) $\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$

Unit matrix (identity matrix) :

Unit matrix is a diagonal matrix in which all the diagonal elements are unity. Unit matrix of order 'n' is denoted by I_n (or I).

i.e. $A = [a_{ij}]_n$ is a unit matrix when $a_{ij} = 0$ for $i \neq j$ & $a_{ii} = 1$

eg. $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Triangular Matrix :

(a) **Upper Triangular Matrix :** A Square Matrix $[a_{ij}]$ is called the upper triangular Matrix, if $a_{ij} = 0$ when $i > j$.

eg. $\begin{bmatrix} 3 & 1 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & 6 \end{bmatrix}$ is a upper triangular matrix of order 3 × 3

(b) **Lower Triangular Matrix :** A Square Matrix $[a_{ij}]$ is called the lower Triangular Matrix, if

$a_{ij} = 0$ when $i < j$

eg. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 2 \end{bmatrix}$ is a lower triangular matrix of order 3 × 3.

Q.1 Find number of zeroes in an identity matrix of order n.

Ans. $n^2 - n$

Sol. No. of zeroes = Total no. of elements – no. of diagonal elements

$= n^2 - n$

Q.2 Find minimum number of zereos in an upper triangular matrix of order n.

Ans. $\frac{n^2 - n}{2}$

Sol. In case of minimum no. of zeroes only elements situated on lower side of principle diagonal will be zero.

min. no. of zeroes = no. of elements in L.T.

$= \frac{(\text{Total elements} - \text{diagonal elements})}{2}$

$= \frac{n^2 - n}{2}$

Q.3 Find maximum number of distinct elements in an upper triangular matrix of order n.

Ans. $\frac{n^2 + n + 2}{2}$

Sol. Maximum number of distinct elements = no. of elements in U.T. + no. of elements in diagonal + 1 (for zeroes)

$= \frac{n^2 - n}{2} + n + 1 = \frac{n^2 + n + 2}{2}$

Q.4 Find maximum number of distinct elements in a diagonal matrix of order n.

Ans. n + 1

Sol. Maximum no. of distinct eleements in diagonal matrix = no. of elements in a diagonal + 1
= n + 1

2.1 EQUALITY OF TWO MATRICES

Two matrices A and B are said to be equal if they are comparable and all the corresponding elements are equal.

Let $A = [a_{ij}]_{m \times n}$ & $B = [b_{ij}]_{p \times q}$

$A = B$ iff (i) $m = p, n = q$

(ii) $a_{ij} = b_{ij} \ \forall$ i & j.

Q.1 Find the value of x, y, z and w which satisfy the matrix equation

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-8 \end{bmatrix} = \begin{bmatrix} -x-1 & 0 \\ 3 & 2w \end{bmatrix}$$

Ans x = – 2, y = 1, z = 4, w = 4

Sol. As the given matrices are equal so their corresponding elements are equal.

$x + 3 = -x - 1 \Rightarrow 2x = -4$

$\therefore x = -2$ .....(i)

$2y + x = 0 \Rightarrow 2y - 2 = 0$ [from (i)]

$\Rightarrow y = 1$ .....(ii)

$z - 1 = 3 \Rightarrow z = 4$ .....(iii)

$4w - 8 = 2w \Rightarrow 2w = 8$

$\therefore w = 4$ .....(iv) **Ans.**

2.2 ALGEBRA OF MATRICES - ADDITION & SUBTRACTION

If A $[a_{ij}]_{m \times n}$ and $[b_{ij}]_{m \times n}$ are two matrices of the same order then their sum A + B is a matrix whose each element is the sum of corresponding element.

i.e. $A + B = [a_{ij} + b_{ij}]_{m \times n}$

Similarly their subtraction A – B is defined as $A - B = [a_{ij} - b_{ij}]_{m \times n}$

eg. If $A = \begin{bmatrix} 5 & 2 \\ 1 & 3 \\ 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}$

then $A + B = \begin{bmatrix} 5+1 & 2+5 \\ 1+2 & 3+2 \\ 4+3 & 1+3 \end{bmatrix} = \begin{bmatrix} 6 & 7 \\ 3 & 5 \\ 7 & 4 \end{bmatrix}$ $\quad A - B = \begin{bmatrix} 5-1 & 2-5 \\ 1-2 & 3-2 \\ 4-3 & 1-3 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 1 \\ 1 & -2 \end{bmatrix}$

Note :

Matrix addition and subtraction can be possible only when Matrices are of same order.

Properties of Matrices addition :

If A, B and C are Matrices of same order, then-

1. $A + B = B + A$
2. $(A+ B) + C = A + (B + C)$
3. $A + O = O + A = A$; $A - O = A$ where O is zero matrix
4. If $A + B = O$ then A and B are called additive inverse of each other
5. $tr\ (A \pm B) = tr\ (A) \pm tr\ (B)$
6. $|A + B| \neq |A| + |B|$

2.3 MULTIPLICATION OF A MATRIX BY A SCALAR

If $A = [a_{ij}]_{m \times n}$ then $kA = [ka_{ij}]_{m \times n}$ $\quad k \in R$

If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$; $kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$

Properties of scalar multiplication :

1. $k(A + B) = kA + kB.$
2. $(k_1 + k_2)A = k_1A + k_2A.$
3. $(k_1k_2)A = k_1(k_2A) = k_2(k_1A)$
4. $tr(kA) = k\ tr(A)$
5. $|kA| = k^n|A|$

Q.1 $A = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix}$ Find (i) 3A (ii) 5A (iii) $-A$

Ans. (i) $3A = \begin{bmatrix} 9 & 6 \\ 0 & 15 \end{bmatrix}$ (ii) $5A = \begin{bmatrix} 15 & 10 \\ 0 & 25 \end{bmatrix}$ (iii) $-A = \begin{bmatrix} -3 & -2 \\ 0 & -5 \end{bmatrix}$

Q.2 If $A = \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ 3 & 2 \end{bmatrix}$ Find (i) $A + B$ (ii) $A - B$ (iii) $2A + 3B - 5I$

Ans. (i) $A+B=\begin{bmatrix}8 & 3\\ 5 & 2\end{bmatrix}$ (ii) $A-B=\begin{bmatrix}-2 & -1\\ -1 & -2\end{bmatrix}$ (iii) $2A+3B-5I=\begin{bmatrix}16 & 8\\ 13 & 1\end{bmatrix}$

Sol. (i) $A+B=\begin{bmatrix}3+5 & 1+2\\ 2+3 & 0+2\end{bmatrix}=\begin{bmatrix}8 & 3\\ 5 & 2\end{bmatrix}$

(ii) $A-B=\begin{bmatrix}3-5 & 1-2\\ 2-3 & 0-2\end{bmatrix}=\begin{bmatrix}-2 & -1\\ -1 & -2\end{bmatrix}$

(iii) $2A+3B-5I=\begin{bmatrix}6 & 2\\ 4 & 0\end{bmatrix}+\begin{bmatrix}15 & 6\\ 9 & 6\end{bmatrix}-\begin{bmatrix}5 & 0\\ 0 & 5\end{bmatrix}$

$=\begin{bmatrix}6+15-5 & 2+6-0\\ 4+9-0 & 0+6-5\end{bmatrix}=\begin{bmatrix}16 & 8\\ 13 & 1\end{bmatrix}$

3.1 MULTIPLICATION OF MATRICES

Let A be a matrix of order m × n and B be a matrix of order p × q, then the matrix multiplication AB is possible if and only if n = p and matrices are said to be **conformable** for multiplication.

In the product AB, A is called pre-factor and B is called post factor.

$\Rightarrow$ AB is possible if and only if number of columns in pre-factor = number of rows in post-factor.

Let $A_{m\times n}=[a_{ij}]$ and $B_{n\times p}=[b_{ij}]$ then order of AB is m × p & $\boxed{(AB)_{ij}=\sum_{r=1}^{n}a_{ir}b_{rj}}$

e.g. $A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\end{bmatrix}_{2\times 3}$ and $B=\begin{bmatrix}b_{11} & b_{12} & b_{13} & b_{14}\\ b_{21} & b_{22} & b_{23} & b_{24}\\ b_{31} & b_{32} & b_{33} & b_{34}\end{bmatrix}_{3\times 4}$

Then order of AB is 2 × 4.

$(AB)_{11}=a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31}=\sum_{r=1}^{3}a_{1r}b_{r1}$

$(AB)_{23}=a_{21}b_{13}+a_{22}b_{23}+a_{23}b_{33}=\sum_{r=1}^{3}a_{2r}b_{r3}$

In general $(ab)_{ij}=a_{i1}b_{1j}+a_{i2}b_{2j}+a_{i3}b_{3j}=\sum_{r=1}^{3}a_{ir}b_{rj}$

Properties of Matrix Multiplication :

If A, B and C are three matrices such that their product is defined , then

(i) $AB \neq BA$

(ii) (AB) C = A (BC)

(iii) IA = A = AI I is identity matrix for matrix multiplication

(iv) A (B + C) = AB + AC

(v) AO = O = OA

(vi) tr(AB) = tr(BA)

(vii) If AB = BA, then A & B commute with each other.

If AB = –BA, then A & B anti commute with each other.

(viii) $|AB| = |A||B|$

Note

1. If A and B are two matrices of the same order, then

(a) $(A + B)^2 = A^2 + B^2 + AB + BA$

(b) $(A - B)^2 = A^2 + B^2 - AB - BA$

(c) $(A - B)(A + B) = A^2 - B^2 + AB - BA$

2. Positive Integral Powers of A square matrix :

For a square matrix A, $A^n = \underbrace{A.A.A.....A}_{\text{upto n times}}$ where $n \in N$

(i) $A^m.A^n = A^{m+n}$ (ii) $(A^m)^n = A^{mn}$, where $m,n \in N$

(ii) If A and B are square matrices of same order and AB = BA then

$(A + B)^n = {}^nC_0A^n + {}^nC_1A^{n-1}B + {}^nC_2A^{n-2}B^2 + ... + {}^nC_nB^n$

Note that for a unit matrix I of any order , $I^m = I$ for all $m \in N$.

Q.1 $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 5 & 5 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Find (i) AB (ii) BC

Ans. (i) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Sol. (i) $AB = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 5 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0\times5+1\times0 & 0\times5+1\times0 \\ 0\times5+2\times0 & 0\times5+2\times0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(ii) $BC = \begin{bmatrix} 5 & 5 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 5\times0+5\times0 & 5\times0+5\times0 \\ 0\times0+0\times0 & 0\times0+0\times0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Q.2 $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$ Find (i) $A^2 + AB$ (ii) $AB + BA$

Ans. (i) $A^2 + AB = \begin{bmatrix} 22 & 17 \\ 10 & 11 \end{bmatrix}$ (ii) $AB + BA = \begin{bmatrix} 14 & 9 \\ 16 & 18 \end{bmatrix}$

Sol. (i) $A^2 + AB = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$

$= \begin{bmatrix} 10 & 9 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 12 & 8 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 22 & 17 \\ 10 & 11 \end{bmatrix}$

(ii) $AB + BA = \begin{bmatrix} 12 & 8 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

$= \begin{bmatrix} 12 & 8 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 12 & 14 \end{bmatrix} = \begin{bmatrix} 14 & 9 \\ 16 & 18 \end{bmatrix}$

Q.3 Prove that diagonal matrices of same order commutes with each ether.

Sol. $A = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$ $B = \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix}$

$AB = \begin{bmatrix} a_1b_1 & 0 \\ 0 & a_2b_2 \end{bmatrix}$ $BA = \begin{bmatrix} a_1b_1 & 0 \\ 0 & a_2b_2 \end{bmatrix}$

$\Rightarrow AB = BA$

Q.4 Find matrix A which commutes with B when $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Ans. $A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$

Sol. $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$

$AB = BA$

$\begin{bmatrix} a & c \\ b & d \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix}$

$\begin{bmatrix} a & a+c \\ b & b+d \end{bmatrix} = \begin{bmatrix} b+d & c+d \\ b & d \end{bmatrix}$

$a = b + d$ $a + c = c + d$ $b + d = d$

$a = d$ $a = d$ $b = 0$

$B = \begin{bmatrix} d & c \\ 0 & d \end{bmatrix} = \begin{bmatrix} a & c \\ 0 & a \end{bmatrix}$

Q.5 $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ If $(A + B)^2 = A^2 + B^2$ Find a and b.

Ans $a = 1; b = 4$

Sol. $A^2 + AB + BA + B^2 = A^2 + B^2$

$AB + BA = 0$

$$\begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} + \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} = 0$$

$$\begin{bmatrix} a-b & 1+1 \\ 2a-b & 2+1 \end{bmatrix} + \begin{bmatrix} a+2 & -a-1 \\ b-2 & -b+1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2a-b+2 & 1-a \\ 2a+b-2 & 4-b \end{bmatrix} = 0$$

$b = 4 \qquad a = 1$

Q.6 If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ prove that $A^2 - 4A + 7I = 0$

Sol. $$A^2 - 4A + 7I = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 4\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Q.7 Find an upper triangular matrix A such that $A^3 = \begin{bmatrix} 8 & -76 \\ 0 & 27 \end{bmatrix}$

Ans $A = \begin{bmatrix} 2 & -4 \\ 0 & 3 \end{bmatrix}$

Sol. Let $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} a^2 & ab+bc \\ 0 & c^2 \end{bmatrix}$

$$A^2 \times A = \begin{bmatrix} a^3 & a^2b + abc + bc^2 \\ 0 & c^3 \end{bmatrix}$$

$a^3 = 8 \qquad c^3 = 27$

$a = 2 \qquad c = 3$

$4b + 6b + 9b = -76$

$19b = -76$

$b = -4$

$$A = \begin{bmatrix} 2 & -4 \\ 0 & 3 \end{bmatrix}$$

Q.8 If $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ If $A^2 + pA + qI = 0$ Find p & q.

Ans $p = -5 ; q = 4$

Sol. $A^2 + pA + qI = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} + p\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} + q\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 11 & 10 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 3p & 2p \\ p & 2p \end{bmatrix} + \begin{bmatrix} q & 0 \\ 0 & q \end{bmatrix}$

$\begin{bmatrix} 11+3p+q & 10+2p \\ 5+p & 6+2p+q \end{bmatrix} = 0$

$5 + p = 0$ $\qquad\qquad$ $6 + 2p + q = 0$

$p = -5$ $\qquad\qquad$ $6 - 10 + q = 0$

$q = 4$

Q.9 If $AB = O$, $BC = I$ Prove that $(A + B)^2 (A + C)^2 = I$.

Sol. $AB = 0$

$A \times (BC = I)$

$(AB)C = AI$

$(0)C = A$

$A = 0$

$(A + B)^2 (A + C)^2 = (0 + B)^2 (0 + C)^2 = B^2C^2$

$= BBCC = BIC = BC = I$

Note:

1. $M\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \ \\ \ \end{bmatrix} \Rightarrow$ Gives I[st] column; $\qquad M\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \ \\ \ \end{bmatrix} \Rightarrow$ Gives II[nd] column

$M\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \ \\ \ \end{bmatrix} \Rightarrow$ Gives III[rd] column

3.2 CHARACTERISTIC EQUATION

Matrix Polynomial :

If $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots\ldots + a_nx^0$, then we define a matrix polynomial

$f(A) = a_0A^n + a_1A^{n-1} + a_2A^{n-2} + \ldots\ldots + a_nI^n$.

where A is the given square matrix. If $f(A)$ is the null matrix, then A is called the zero or root of the polynomial $f(x)$.

Characteristic equation :

Let A be a square matrix. Then the polynomial $|A - xI|$ is called as characteristic polynomial of A & the equation $|A - xI| = 0$ is called as characteristic equation of A. After solving the characterstic polynomial the values of 'x' are said to be characterstic roots of the polynomial.

Roots of characteristic equations are called eigen values of A

Note : (i) Sum of the roots of the characteristic equation is equal to trace of the matrix.

(ii) Product of the roots of the characteristic equation is equal to the determinant value.

(iii) The degree of characteristic equation is same as the order of the matrix.

Find characteristic equation of matrix $A=\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$

$\because\ |A - xI| = 0 \Rightarrow \begin{vmatrix}2-x & 1\\ -1 & 1-x\end{vmatrix} = 0$

$\Rightarrow (2 - x)(1 - x) + 1 = 0 \ \Rightarrow\ x^2 - 3x + 3 = 0$ (characterstic polynomial)

Important Points

A is a square matrix of order n. $|A - xI| = 0$ is its characteristic equation. Roots of characteristic equations are called eigen values of A. Let say eigen values are $x_1, x_2, ...x_n$

1. If f(x) is a polynomial, so for matrix f(A), its eigen values will be $f(x_1), f(x_2), ...f(x_n)$
   So we can directly write its characteristic equation $(x - f(x_1))(x - f(x_2)).....(x - f(x_n))=0$
   For example
   Let A is a square matrix of order 2 and its characteristic equation is $x^2 - x - 2 = 0$ So its eigen values will be 2 and -1. If $f(x) = x^3 + x$, so for matrix $A^3 + A$, its eigen values will be 10, -2 and characteristic equation will be $(x - 10)(x + 2) = 0$
2. Eigen values of A and A^T are same
3. If eigen values of A are x_i then eigen values of A-1 are $1/x_i$
4. If A is triangular then diagonal elements are eigen values of A

Q.1 Find characteristic equation of matrix A.

(i) $A=\begin{bmatrix}2 & 3\\ 1 & 5\end{bmatrix}$ (ii) $A=\begin{bmatrix}2 & 0 & 1\\ 0 & 1 & 2\\ 5 & 0 & 3\end{bmatrix}$

Ans. (i) $x^2 - 7x + 7 = 0$ (ii) $x^3 - 6x^2 + 6x - 1 = 0$

Sol. (i) $A - \lambda I = \begin{bmatrix}2 & 3\\ 1 & 5\end{bmatrix} - \lambda\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$A - \lambda I = \begin{bmatrix}2-\lambda & 3\\ 1 & 5-\lambda\end{bmatrix}$

$|A - \lambda I| = 0$

$\begin{vmatrix}2-\lambda & 3\\ 1 & 5-\lambda\end{vmatrix} = 0$

$(2 - \lambda)(5 - \lambda) - 3 = 0$

$\lambda^2 - 7\lambda + 7 = 0$

$A^2 - 7A + 7I = 0$

sum of roots = tr(A)

Product of roots = |A|

(ii) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \\ 5 & 0 & 3 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \\ 5 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 1-\lambda & 2 \\ 5 & 0 & 3-\lambda \end{bmatrix}$$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 1-\lambda & 2 \\ 5 & 0 & 3-\lambda \end{vmatrix} = 0$$

$(2 - \lambda)(\lambda^2 - 4\lambda + 3) - 1(5(1 - \lambda)) = 0$
$-\lambda^3 + 4\lambda^2 - 3\lambda + 2\lambda^2 - 8\lambda + 6 - 5 + 5\lambda = 0$
$-\lambda^3 + 6\lambda^2 - 6\lambda + 1 = 0$

Q.2 If $A = \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix}$; Find $|A^4 - 12A^3 + 44A^2 - 48A + 21I|$

Ans 25

Sol. $A^2 - 6A + 4I = 0$
$x^2 - 6x + 4 = 0$

$$\begin{array}{l} x^2-6x+4 \overline{)\ x^4 - 12x^3 + 44x^2 - 48x + 21\ (} \\ \quad x^4 - 6x^3 + 4x^2 \\ \quad - \quad + \quad - \\ \hline \quad -6x^3 + 40x^2 - 48x + 21 \\ \quad -6x^3 + 36x^2 - 24x \\ \quad + \quad - \quad + \\ \hline \quad 4x^2 - 24x + 21 \\ \quad 4x^2 - 24x + 16 \\ \quad - \quad + \quad - \\ \hline \quad 5 \end{array}$$

$|A^2 - 12A^3 + 44A^2 - 48A + 21I| = (A^2 - 6A + 4I)(A^2 - 6A + 4I)$
$= |(0)(0) + 5I|$
$= |5I| = 5^2|I| = 25$

Q.3 If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$; Find A^5

Ans $A^5 = -31A - 56I$

Sol. Characteristics equation $A^2 - 4A + 7I = 0$

$$
\begin{array}{r}
x^3 + 4x^2 + 9x + 8 \\
x^2 - 4x + 7 \overline{)\ x^5 \qquad\qquad\qquad\qquad} \\
\underline{x^5 - 4x^4 + 7x^3} \\
4x^4 - 7x^3 \\
\underline{4x^4 - 16x^3 + 28x^2} \\
9x^2 - 28x^2 \\
\underline{9x^3 - 36x^2 + 63x} \\
8x^2 - 63x \\
\underline{8x^2 - 32x + 56} \\
-31x - 56
\end{array}
$$

$$A^5 = \underbrace{\left(A^3 + 9A^2 + 8I\right)\left(A^2 - 4A + 7I\right)}_{0} - 31A - 56I$$

$= 0 - 31A - 56I$

$A^5 = -31A - 56I$

Method II :

$A^2 = 4A - 7I$

$A^4 = (4A - 7I)^2$

$= 16A^2 - 56A + 49I = 16(4A - 7I) - 56A + 49I$

$A^4 = 8A - 63I$

$\times$ by A

$A^5 = 8A^2 - 63A = 8(4A - 7I) - 63A$

$= 32A - 56I - 63A = -31A - 56I$

Q.4 Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that $AB = B$ and $a + d = 2021$, then the value of $ad - bc$ is equal to _____.

[JEE Main Mar. 2021]

माना $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ तथा $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ हैं, जिनके लिए $AB = B$ तथा $a + d = 2021$ है, तो $ad - bc$ का मान बराबर है _____

[JEE Main Mar. 2021]

Ans. 2020

Sol. $AB = B$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\begin{bmatrix} a\alpha + & b\beta \\ c\alpha + & d\beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$a\alpha + b\beta = \gamma$

$c\alpha + d\beta = \beta$

$(a-1)\alpha + b\beta = 0$

$c\alpha + (d-1)\beta = 0$

here $\alpha = \beta \neq 0$

then $\begin{vmatrix} a-1 & b \\ c & d-1 \end{vmatrix} = 0$

$(a-1)(d-1) - bc = 0$

$ad - a - d + 1 - bc = 0$

$ad - bc = (a + d) - 1$

$ad - bc = 2021 - 1$

$= 2020$

Q.5 Let I be an identity matrix of order 2×2 and $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of $n \in N$ for which $P^n = 5I - 8P$ is equal to _____. **[JEE Main Mar. 2021]**

माना I, कोटि 2×2 का तत्समक आव्यूह है तथा $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$ है। तो $n \in N$ का वह मान, जिसके लिए $P^n = 5I - 8P$ है, बराबर _____ **[JEE Main Mar. 2021]**

Ans. 6

Sol. $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$

$$P^2 = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ -5 & 4 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} -1 & 1 \\ -5 & 4 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix}$$

$$5I - 8P = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 16 & -8 \\ 40 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix}$$

$$P^6 = \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix}\begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix}$$

$n = 6$

4. IMPORTANT DEFINITONS

1. Idempotent Matrix : A square matrix is idempotent provided $A^2 = A$.

For idempotent matrix note the following :

(i) $A^n = A \ \forall \ n \geq 2, n \in N$.

(ii) determinant value of idempotent matrix is either 0 or 1

2. Periodic Matrix : A square matrix which satisfies the relation $A^{k+1} = A$, for some positive integer K, is a periodic matrix. The period of the matrix is the least value of K for which this holds true.

Note that period of an idempotent matrix is 1.

3. Nilpotent Matrix : A square matrix of the order 'n' is said to be nilpotent matrix of order m, $m \in N$, if $A^m = \mathbf{O}$ & $A^{m-1} \neq \mathbf{O}$.

Eg.: If $A^3 = 0$ & $A^2 \neq 0$

$(I + A)^n = I + {}^nC_1A + {}^nC_2A^2 + ({}^nC_3A^3 + \ldots\ldots {}^nC_nA^n)$

Q.1 $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, find B^{50}.

Note:

1. Index of nilpotency of a square matrix will not be greater than its order
2. Any n-order triangular matrix with zeros along the main diagonal is nilpotent, with index $\leq$ n

3. Any matrix of the form $\begin{bmatrix} a & -a \\ a & -a \end{bmatrix}$, $\begin{bmatrix} a & a & a \\ b & b & b \\ -a-b & -a-b & -a-b \end{bmatrix}$ is nilpotent

4. Another examples are $\begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$, $\begin{bmatrix} 2 & 2 & -2 \\ 5 & 1 & -3 \\ 1 & 5 & -3 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ $A^n = \begin{bmatrix} 1 & na & nb + {}^nC_2 (ac) \\ 0 & 1 & nc \\ 0 & 0 & 1 \end{bmatrix}$

4. Involutary Matrix : If $A^2 = I$, the matrix is said to be an involutary matrix. i.e. square roots of identity matrix is involutary matrix.

Note : The determinant value of involutary matrix is 1 or –1.

5. TRANSPOSE OF A MATRIX

The matrix obtained from a given matrix A by changing its rows into columns or columns into rows is called transpose of Matrix A and is denoted by A^T or A'.

From the definition it is obvious that

If order of A is m × n, then order of A^T is n × m.

eg. Transpose of Matrix $\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}_{2 \times 3}$ is $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}_{3 \times 2}$

Properties of Transpose :

(i) $\left(A^T\right)^T = A$

(ii) $(A \pm B)^T = A^T \pm B^T$

(iii) $(AB)^T = B^T A^T$

(iv) $(kA)^T = k(A)^T$

(v) $(A_1 A_2 A_3 \ldots\ldots A_{n-1} A_n)^T = A_n^T A_{n-1}^T \ldots\ldots A_3^T A_2^T A_1^T$

(vi) tr (A) = tr (A^T) (vii) tr(AB) = tr(BA)

6. SYMMETRIC AND SKEW-SYMMETRIC MATRIX

(a) Symmetric matrix :

A square matrix $A = [a_{ij}]$ is said to be, symmetric if, $a_{ij} = a_{ji}$ $\forall$ i & j (conjugate elements are equal).

Hence for symmetric matrix $\mathbf{A = A^T}$.

Note : Max. number of distinct entries in any symmetric matrix of order n is $\frac{n(n+1)}{2}$.

(b) Skew symmetric matrix :

Square matrix $A=[a_{ij}]$ is said to be skew symmetric if $a_{ij}=-a_{ji}\ \forall$ i & j (the pair of conjugate elements are additive inverse of each other). For a skew symmetric matrix $\mathbf{A=-A^T}$.

Note :

(i) If A is skew symmetric, then $a_{ii}=-a_{ii} \Rightarrow a_{ii}=0\ \forall$ i. Thus the diagonal elements of a skew square matrix are all zero, but not the converse.

(ii) The determinant value of odd order skew symmetric matrix is zero.

(iii) Every square matrix can be uniquely expressed as a sum or difference of a symmetric and a skew symmetric matrix.

$$A=\underbrace{\frac{1}{2}(A+A^T)}_{\text{symmetric}}+\underbrace{\frac{1}{2}(A-A^T)}_{\text{skew symmetric}} \text{ and } A=\frac{1}{2}(A^T+A)-\frac{1}{2}(A^T-A)$$

Q.1 If $A=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$ Find A^T

Ans. $A^T=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$

Q.2 Prove that AA^T is symmetric.

Sol. $(AA^T)^T=(A^T)^T(A)^T=AA^T$

$\Rightarrow AA^T$ is symmetric.

Q.3 Prove that $A-A^T$ is skew symmetric

Sol. $(A-A^T)^T=(A^T)-(A^T)^T$

$=A^T-A=-(A-A^T)$

$\Rightarrow A-A^T$ is skew-symmetric.

Q.4 If A is symmetric as well as skew-symmetric then Prove that A will be null matrix.

Sol. Symmetric : $\begin{bmatrix}a & d & e\\ d & b & f\\ e & f & c\end{bmatrix}$

$A=A^T$ (symmetric)

$A=-A^T$ (skew-symmetric)

$2A=0$

$A=0$

If A is a skew-symmetric matrix of odd order then A will be singular.

$$A=\begin{bmatrix}0 & a & b\\ -a & 0 & c\\ -b & -c & 0\end{bmatrix}$$

$A=-A^T$

$|A|=|-A^T|$

$|A| = (-1)^n |A^T|$ ($\because$ n is odd)

$|A| = -|A|$

$\Rightarrow |A| = 0$

Q.5 Find maximum number of distinct entries in a symmetric matrix of order n.

Ans. $\frac{n^2+n}{2}$

Sol. $= \frac{n^2-n+n}{2}$

$$\begin{bmatrix} & & \frac{n^2-n}{2} \\ & n & \\ \frac{n^2-n}{2} & & \end{bmatrix}$$

$= \frac{n^2+n}{2}$

All the entries in UT & LT will be same. So, no. of maximum distinct entries = diag. + U.T.

Q.6 Find maximum number of distinct intries in a skew-symmetric matrix of order n.

Ans. n^2+n+1

Sol. $\frac{n^2-n}{2} + \frac{n^2-n}{2} + 1$

$$\begin{bmatrix} & & \frac{n^2-n}{2} \\ & n & \\ \frac{n^2-n}{2} & & \end{bmatrix}$$

$n^2 - n + 1$

Q.7 Prove that $A + A^T$ is symmetric

Sol. $(A+A^T)^T = A^T + (A^T)^T = A + A^T$

$\Rightarrow A + A^T$ is symmetric.

Q.8 Prove that A^TA is symmetric.

Sol. $(A^TA)^T = A^T \cdot (A^T)^T = A^TA$

$\Rightarrow A^TA$ is symmetric.

Q.9 If A and B are symmetric then

(i) Prove that A + B is symmetric

(ii) Prove that AB + BA is symmetric

(iii) Prove that AB – BA is skew-symmetric

(iv) Prove that A^n is symmetric $\forall n \in N$

(v) Prove that C^TAC is symmetric

Sol. (i) $A = A^T$

$B = B^T$

$(A + B)^T = A^T + B^T = A + B$

$\Rightarrow A + B$ is symmetric.

(ii) $(AB + BA)^T = (AB)^T + (BA)^T$

$= B^TA^T + A^TB^T$

$= BA + AB$

$\Rightarrow AB + BA$ is symmetric.

(iii) $(AB - BA)^T = (AB)^T - (BA)^T$

$= B^TA^T - A^TB^T$

$= BA - AB = -(AB - BA)$

$\Rightarrow AB - BA$ is skew-symmetric.

(iv) $(A^n)^T = (A^T)^n = A^n$

$\Rightarrow A^n$ is symmetric.

(v) $(C^TAC)^T = (C)^T (A)^T (C^T)^T$

$= C^TA^TC = C^TAC$

$\Rightarrow C^TAC$ is symmetric.

Q.10 If A is skew-symmetric then

(i) Prove that B^TAB is skew-symmetric (ii) Prove that A^n is skew-symmetric if n is odd

(iii) Prove that A^n is symmetric if n is even

Sol. (i) $(B^TAB)^T = (B)^T (A)^T (B^T)^T = B^TA^TB$

$A^T = -A$ (given)

$= -B^TAB$

(ii) $(A^{odd})^T = (A^T)^{odd} = (-A)^{odd} = -A^{odd}$

(iii) $(A^{even})^T = (A^T)^{even} = A^{even}$

$\Rightarrow A^{even}$ is symmetric.

Q.11 $A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & y-x & -1 \\ x & 1 & 2 \end{bmatrix}$ If AB is symmetric find x and y.

Ans. $x = -2, y = 1$

Sol. $A^T = \begin{bmatrix} 1 & 3 & -1 \\ -2 & 1 & 2 \end{bmatrix}$

$B = \begin{bmatrix} 1 & y-x & -1 \\ x & 1 & 2 \end{bmatrix}$

$\Rightarrow x = -2$ $y - x = 3$ $y = 1$

M – II

AB = BA (AB is symmetric)

Q.12 $A = \begin{bmatrix} 5 & 2 & x \\ y & z & -3 \\ 4 & t & -7 \end{bmatrix}$ Find x, y, z, t if A is symmetric.

Ans. $x = 4, y = 2, t = -3, z \in R$

Sol. $A_T = \begin{bmatrix} 5 & y & 4 \\ 2 & z & t \\ x & -3 & -7 \end{bmatrix} = A$

$A_T = A$

$\Rightarrow y = 2$ $x = 4$ $t = -3$ $z \in R$

Q.13 Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13}B^{26} - B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13} - C^{13}A^{26}$ is symmetric Then

माना 3×3 के आव्यूहों A, B, C में A सममित है तथा B और C विषम सममित है। कथनों

(S1) $A^{13}B^{26} - B^{26}A^{13}$ सममित है

(S2) $A^{26}C^{13} - C^{13}A^{26}$ सममित है का विचार कीजिए। तो

Ans. S_2 is true and S_1 is false

Sol. $S_1 : (A^{13}B^{26} - B^{26}A^{13})^T = (A^{13}B^{26})T - (B^{26}A^{13})^T$

$\Rightarrow (B^{26})^T (A^{13})^T - (A^{13})^T (B^{26})^T$

$\Rightarrow (BT)^{26} (A^T)^{13} - (A^T)^{13} (B^T)^{26}$

$\Rightarrow (-B)^{26} A^{13} - A^{13} (-B)26$

$\Rightarrow -(A^{13}B^{26} - B26\,A^{13})$ so skew symmetric

$S_2 : (A^{26}C^{13} - C^{13}A^{26})^T$

$= (C^T)^{13} (A^T)^{26} - (A^T)^{26} (C^T)^{13}$

$= - C^{13}A^{26} + A^{26} . C^{13}$

So, symmetric matrix

Hence S_2 is true and S_1 is false

7. ORTHOGONAL MATRIX

A square matrix is said to be orthogonal matrix if $AA^T = I = A^TA$

Note :

(i) The determinant value of orthogonal matrix is either 1 or –1.

(ii) Three rows of an orthogonal matrix are components of three mutually perpendicular unit vectors.

Proof : Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ $\Rightarrow A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

$$AA^T = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

If $AA^T = I$, then

$\sum_{i=1}^{3} a_i^2 = \sum b_i^2 = \sum c_i^2 = 1$ and $\sum_{i=1}^{3} a_ib_i = \sum_{i=1}^{3} b_ic_i = \sum_{i=1}^{3} c_ia_i = 0$

Q.1 Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of M^TM is seven, is _____ . **[JEE Main Feb. 2021]**

माना M कोई 3×3 आव्यूह है जिसके अवयव समुच्चय $\{0, 1, 2\}$ से लिए गए हैं। इस तरह के आव्यूहों की अधिकतम संख्या, जिनके

लिए M^TM के विकर्ण के अवयवों का योग 7 है, है _____ ।

Ans. 540

Sol. $M=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$tr(M^TM) = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 7$

Case – 1 2, 1, 1, 1, 0, 0, 0, 0, 0

Case – 2 1, 1, 1, 1, 1, 1, 1, 0, 0

$$\text{Total} = \frac{9!}{3!5!} + \frac{9!}{7!2!}$$

$= 540$

Q.2 Let $P=\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$. If $P^TQ^{2007}\ P=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b – 3c – 4d$ equal to :

माना $P=\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ तथा $Q = PAP^T$ है। यदि $P^TQ^{2007}\ P=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, तो $2a + b – 3c – 4d$ बराबर है :

Ans. 2005

Sol. $PP^T=\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}=I = P^TP$

$P^TQ^{2007}P = P^T\left[\left(PAP^T\right)\left(PAP^T\right).....\left(PAP^T\right)\right]P$

$= P^T\left[PA^{2007}P^T\right]P$

$= A^{2007}$

$A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$A^2=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$A^3=\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

$$A^{2007} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$a = 1, b = 2007, c = 0, d = 1$

$2a + b - 3c - 4d = 2005$

Note:

$$A(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, A(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$

(i) $(A(\theta))^n = A(n\theta)$ (ii) $AA^T = I$ (iii) $A(\alpha)\, A(\beta) = A(\alpha + \beta)$

8. ADJOINT OF A SQUARE MATRIX

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A, denoted by adjA, is defined as the transpose of the cofactor matrix.

Then, $\text{adj}A = [C_{ij}]^T \Rightarrow \text{adj}A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{23} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$

Method to find adjoint of a 2 × 2 square matrix, directly :

Let A be a 2 × 2 square matrix. In order to find the adjoint simply interchange the diagonal elements and reverse the sign of off diagonal elements (rest of the elements).

e.g. If $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \text{adj}A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$

Theorem : A (adj. A) = (adj. A) .A = |A| I_n.

Proof : $A.(\text{adj } A) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$

$$\begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \Rightarrow \quad A.(\text{Adj. } A) = |A|\, I$$

(whatever may be the value only |A| will come out as a common element)

If $|A| \neq 0$ then $\dfrac{\mathbf{A.\ (adj.\ A)}}{|\mathbf{A}|} = \mathbf{I}$ = unit matrix of the same order as that of A

Properties of adjoint matrix :

If **A** be a square matrix of order **n**, then

(i) $|\text{adj } A| = |A|^{n-1}$ (ii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$,

(iii) $|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2}$,

(iv) $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

adj(ABC) = adj (C) adj (B) adj (A)

adj $(A_1A_2 \ldots\ldots A_n)$ = adj (A_n) adj (A_{n-1}) ..... adj (A_1)

if $A_1 = A_2 = \ldots\ldots\ldots A_n = A$

adj $(A^n) = (\text{adj } A)^n$

(v) adj(KA) = K^{n-1} (adj A) , K is a scalar

(vi) adj $A^T = (\text{adj } A)^T$

Q.1 If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then adj A is equal to -

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$ (B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$ (C) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$ (D) none of these

Ans A

Sol. adj. $A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$ **Ans. (A)**

Q.2 Prove that adjoint of a diagonal matrix is also a diagonal matrix.

Sol. Let $A = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$

$$\text{adj}A = \begin{bmatrix} a_2 & 0 \\ 0 & a_1 \end{bmatrix}^T = \begin{bmatrix} a_2 & 0 \\ 0 & a_1 \end{bmatrix}$$

Or : $A = \begin{bmatrix} k_1 & 0 & 0 \\ 0 & k_2 & 0 \\ 0 & 0 & k_3 \end{bmatrix}$

$$\text{adj}(A) = \begin{bmatrix} k_2k_3 & 0 & 0 \\ 0 & k_1k_3 & 0 \\ 0 & 0 & k_1k_2 \end{bmatrix}^T = \begin{bmatrix} k_2k_3 & 0 & 0 \\ 0 & k_1k_3 & 0 \\ 0 & 0 & k_1k_2 \end{bmatrix}$$

Q.3. If |A| = 3, |B| = 2 Find $|\text{adj } (A^2B)|$ order of A and B is 3.

Ans 324

Sol. $|A^2B|^{3-1} = (|A|^2\ |B|)^2 = |A|^4\ |B|^2 = (3)^4\ (2)^2$

= 324

Q.4 If A is symmetric then prove that adj (A) is also symmetric.

Sol. To prove : $A = A^T$

$(\text{adj}(A))^T = \text{adj}(A)$

L.H.S.= $(\text{adj}(A^T)) = \text{adj}(A)$

9.1 INVERSE OF A SQUARE MATRIX

A square matrix A said to be invertible if and only if it is non-singular (i.e. $|A| \neq 0$) and there exists a matrix B such that, $AB = I = BA$.

B is called the **inverse** (reciprocal) of A and is denoted by $\mathbf{A^{-1}}$. Thus

$A^{-1} = B \Leftrightarrow AB = I = BA$

We have, $A.(\text{adj } A) = |A| I_n$

$A^{-1}.\ A(\text{adj } A) = A^{-1} I_n |A|$

$I_n(\text{adj } A) = A^{-1} |A| I_n \qquad \therefore\ \mathbf{A^{-1} = \frac{(adj\ A)}{|A|}}$

Note : The necessary and sufficient condition for a square matrix A to be invertible is that $\mathbf{|A| \neq 0}$

Q.1 $AB = 5I$ Find (i) A^{-1} (ii) $(3A)^{-1}$

Ans. (i) $A^{-1} = \frac{B}{5}$ (ii) $(3A)^{-1} = \frac{B}{15}$

Sol. (i) A^{-1}

$$A\left(\frac{B}{5}\right) = I$$

$$A^{-1} = \frac{B}{5}$$

(ii) $(3A)^{-1}$

$$(3A)\left(\frac{B}{15}\right) = I$$

$$(3A)^{-1} = \frac{B}{15}$$

Q.2 $\begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Find A^{-1}

Ans. $A^{-1} = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$

Q.3 $A = \begin{bmatrix} k_1 & 0 & 0 \\ 0 & k_2 & 0 \\ 0 & 0 & k_3 \end{bmatrix}$ Find A^{-1}

Ans. $A^{-1}=\begin{bmatrix}\frac{1}{k_1} & 0 & 0\\ 0 & \frac{1}{k_2} & 0\\ 0 & 0 & \frac{1}{k_3}\end{bmatrix}$

Sol. $\text{adj}(A)=\begin{bmatrix}k_2k_3 & 0 & 0\\ 0 & k_1k_3 & 0\\ 0 & 0 & k_1k_2\end{bmatrix}^T=\begin{bmatrix}k_2k_3 & 0 & 0\\ 0 & k_1k_3 & 0\\ 0 & 0 & k_1k_2\end{bmatrix}$

$$A^{-1}=\frac{\begin{bmatrix}k_2k_3 & 0 & 0\\ 0 & k_1k_3 & 0\\ 0 & 0 & k_1k_2\end{bmatrix}}{k_1k_2k_3}=\begin{bmatrix}\frac{1}{k_1} & 0 & 0\\ 0 & \frac{1}{k_2} & 0\\ 0 & 0 & \frac{1}{k_3}\end{bmatrix}$$

Q.4 $A=\begin{bmatrix}2 & 5\\ 1 & 3\end{bmatrix}$ Find A^{-1}

Ans. $A^{-1}=\begin{bmatrix}3 & -5\\ -1 & 2\end{bmatrix}$

Sol. $A^{-1}=\frac{\text{adj}(A)}{|A|}=\frac{\begin{bmatrix}3 & -5\\ -1 & 2\end{bmatrix}}{1}=\begin{bmatrix}3 & -5\\ -1 & 2\end{bmatrix}$

Properties of Inverse :

(i) $AA^{-1}=I=A^{-1}A$

(ii) $(A^{-1})^{-1}=A$

(iii) $(A^k)^{-1}=(A^{-1})^k=A^{-k};\ k\in N$

(iv) $(A^T)^{-1}=(A^{-1})^T$

(v) $|A^{-1}|=\frac{1}{|A|}$

(vi) $(kA)^{-1}=\frac{1}{k}(A^{-1})\ k\in R$

(vii) If A & B are invertible matrices of the same order, then $\mathbf{(AB)^{-1}=B^{-1}A^{-1}}$.

Note: If $A_1, A_2, \ldots\ldots\ldots\ldots A_n$ are all invertible square matrices of order n then $(A_1A_2\ldots\ldots\ldots\ldots A_n)^{-1}=A_n^{-1}A_{n-1}^{-1}\ldots\ldots\ldots\ldots A_2^{-1}A_1^{-1}$

(viii) Let A,B,C be square matrices of the same order 'n'. If A is a non-singular matrix, then

(a) $AB = AC \Rightarrow B = C$ (Left cancellation law)

(b) $BA = CA \Rightarrow B = C$ (Right cancellation law)

Note that these laws hold only if the matrix 'A' is **non-singular** (i.e.$|A| \neq 0$).

(ix) If $AB = O$

(i) $|A| \neq 0 \Rightarrow B = O$

(ii) $B \neq O \Rightarrow |A| = 0$

Q.5 $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} A \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 1 & 2 \end{bmatrix}$ Find A.

Ans. $A = \begin{bmatrix} 32 & -43 \\ -11 & 15 \end{bmatrix}$

Sol. $PAQ = R$

$P^{-1}PAQ = P^{-1}R$

$IAQQ^{-1} = P^{-1}RQ^{-1}$

$AI = P^{-1}Q^{-1}R$

$A = P^{-1}RQ^{-1}$

$P^{-1} = \dfrac{\begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}}{1}$ $\qquad Q^{-1} = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ $\qquad R = \begin{bmatrix} 5 & 3 \\ 1 & 2 \end{bmatrix}$

$P^{-1}RQ^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 5 & 3 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$

$= \begin{bmatrix} 10 & -1 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 32 & -43 \\ -11 & 15 \end{bmatrix}$

Q.6 $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} A = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$ Find A.

Sol. $PA = Q$

$A \neq P^{-1}Q$ Because P^{-1} is not defined.

$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$

$\begin{bmatrix} 2a+4c & 2b+4d \\ a+2c & b+2d \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$

$2a + 4c = 3$

$a + 2c = \dfrac{3}{2}$

$a + 2c = 1$

$\Rightarrow$ A is not defined.

Q.7 If P and Q are different matrices such that $P^3 = Q^3$, $P^2Q = Q^2P$ find $|P^2 + Q^2|$

Ans 0

Sol. $P^3 - P^2Q = Q^3 - Q^2P$

$P^2 (P - Q) = Q^2(Q - P)$

$P^2 (P - Q) - Q^2 (Q - P) = 0$

$P^2 (P - Q) + Q^2 (P - Q) = 0$

$(P^2 + Q^2) (P - Q) = 0$

$\because P - Q \neq 0 \quad \Rightarrow |P^2 + Q^2| = 0$

Q.8 $A = \begin{bmatrix} a & b \\ c & \dfrac{1+bc}{a} \end{bmatrix}$ Prove that $aA^{-1} = (a^2 + bc + 1) I - aA$

Sol. $A^2 - \left(a + \dfrac{1+bc}{a}\right)A + (1 + bc - bc)I = 0$

$aA^2 - (a^2 + bc + 1)A + aI = 0$

$\times$ by A^{-1}

$aA - (a^2 + bc + 1)I + aA^{-1} = 0$

$aA^{-1} = (a^2 + bc + 1)I - aA$

Hence proved.

Q.9 $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Prove that $(f(x))^{-1} = f(-x)$

Sol. L.H.S. $= \left(f(x)\right)^{-1} = \dfrac{\text{adj}\left(f(x)\right)}{|f(x)|}$

$$\text{adj}\left(f(x)\right) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{R.H.S.} = f(-x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

L.H.S. = R.H.S.

Method II :

$f(x)\, f(-x) = I$

$$\begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q.10 If A is orthogonal then prove that A^{-1} is also orthogonal.

Sol. $AA^T = A^TA = I$ (Given)

To prove : $(A^{-1})(A^{-1})^T = I$ $\qquad (A^{-1})^T = (A^T)^{-1}$

$AA^{-1} = I$

$AA^T = I$

$A^T = A^{-1}$ (Given)

To prove

$(A^{-1})^T = (A^{-1})^{-1}$

$(A^T)^T = A \qquad A = A$

Hence proved.

Q.11 If A is symmetric then prove that A^{-1} is also symmetric

Sol. $A^T = A$

To prove $\quad (A^{-1})^T = A^{-1}$

L.H.S. $= (A^{-1})^T = (A^T)^{-1} = A^{-1} =$ R.H.S.

Hence proved.

Q.12 $\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix} A \qquad$ Find A^{-1}

Ans. $A^{-1} = \begin{bmatrix} 1 & 1 \\ 4 & \frac{5}{2} \end{bmatrix}$

9.2 FIND INVERSE BY ELEMENTARY OPERATIONS

Q.1 Find A^{-1} by elementary transformations where $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

Sol. By Row transformation

$A = IA$

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$R_1 \to R_1 \to R_2$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} A$$

$R_1 \to R_1 - R_2$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=\begin{bmatrix}2 & -3\\ -1 & 2\end{bmatrix}A$$

$$\Rightarrow A^{-1}=\begin{bmatrix}2 & -3\\ -1 & 2\end{bmatrix}$$

$$A = AI$$

$$\begin{bmatrix}2 & 3\\ 1 & 2\end{bmatrix}=A\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$C_1 \to \frac{C_1}{2}$$

$$\begin{bmatrix}1 & 3\\ \frac{1}{2} & 2\end{bmatrix}=A\begin{bmatrix}\frac{1}{2} & 0\\ 0 & 1\end{bmatrix}$$

$$C_2 \to C_2 - 3C_1$$

$$\begin{bmatrix}1 & 0\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}=A\begin{bmatrix}\frac{1}{2} & -\frac{3}{2}\\ 0 & 1\end{bmatrix}$$

$$C_1 \to C_1 - C_2$$

$$\begin{bmatrix}1 & 0\\ 0 & \frac{1}{2}\end{bmatrix}=A\begin{bmatrix}2 & -\frac{3}{2}\\ -1 & 1\end{bmatrix}$$

$$C_2 \to 2C_2$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=A\begin{bmatrix}2 & -3\\ -1 & 2\end{bmatrix}$$

$$A^{-1}=\begin{bmatrix}2 & -3\\ -1 & 2\end{bmatrix}$$

10. SYSTEM OF EQUATIONS

Gauss - Jordan method :

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

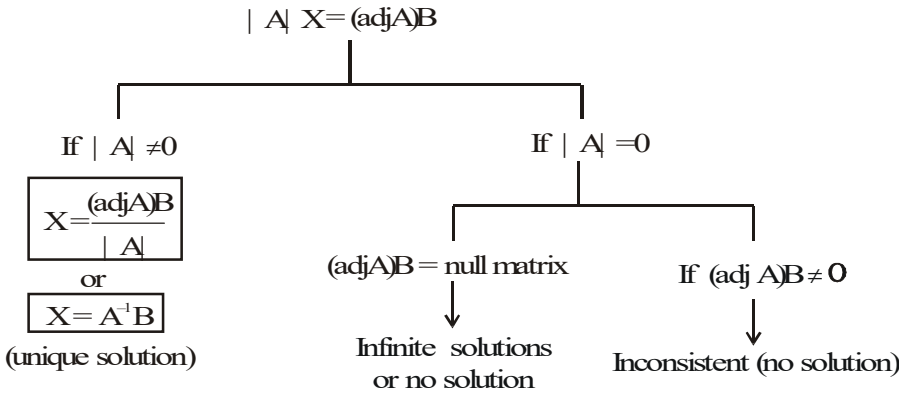
$$a_3x + b_3y + c_3z = d_3$$

$$\Rightarrow \begin{bmatrix}a_1x + b_1y + c_1z\\ a_2x + b_2y + c_2z\\ a_3x + b_3y + c_3z\end{bmatrix}=\begin{bmatrix}d_1\\ d_2\\ d_3\end{bmatrix} \Rightarrow \begin{bmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}d_1\\ d_2\\ d_3\end{bmatrix}$$

$\Rightarrow AX = B$ ....(i)

Multiplying adjA on both the sides of (i)

$\Rightarrow (adjA)\,AX = (adjA)B \quad \Rightarrow |A|X = (adjA)\,B$



11. RANK OF A MATRIX

Rank of a matrix is order of the highest order non-singular sub-matrix.

Ex. $A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ Here $|A| \neq 0$ so, Rank = 3

Q.1 Find the rank of given matrix.

(i) $A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ (iii) $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Ans. (i) 3 (ii) 2 (ii) 1

Note - If rank of a matrix is r then,

(i) all possible sub-matrices of order r + 1 or more will be singular.

(ii) Atleast one sub-matrix of order r will be non-singular.

Sol. (i) $A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

$|A| = 6$

Rank = 3

(ii) $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

$|A| = 0 \qquad \Rightarrow \text{Rank} < 3$

$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

$|A_1| \neq 0 \qquad \text{Rank} = 2$

(iii) $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$A_1 = [1] \qquad |A_1| \neq 0$
$A_2 = [2] \qquad |A_2| \neq 0$
$\Rightarrow \text{Rank} = 1$